Server configuration: https://www.postgresql.org/docs/current/runtime-config.html

https://www.postgresql.org/docs/current/sql-altersystem.html

```
ALTER SYSTEM SET configuration_parameter { TO | = } { value [, ...]
| DEFAULT }

ALTER SYSTEM RESET configuration_parameter
ALTER SYSTEM RESET ALL
```

https://www.postgresql.org/docs/current/sql-alterdatabase.html

```
ALTER DATABASE name [ [ WITH ] option [ ... ] ]

where option can be:

    ALLOW_CONNECTIONS allowconn
    CONNECTION LIMIT connlimit
    IS_TEMPLATE istemplate

ALTER DATABASE name RENAME TO new_name

ALTER DATABASE name OWNER TO { new_owner | CURRENT_ROLE |
CURRENT_USER | SESSION_USER }

ALTER DATABASE name SET TABLESPACE new_tablespace

ALTER DATABASE name REFRESH COLLATION VERSION

ALTER DATABASE name SET configuration_parameter { TO | = } { value
| DEFAULT }
ALTER DATABASE name SET configuration_parameter FROM CURRENT
ALTER DATABASE name RESET configuration_parameter
ALTER DATABASE name RESET ALL
```

```
-- role configurations
ALTER ROLE { role_specification | ALL } [ IN DATABASE
database_name ]
  SET configuration_parameter { TO | = } { value | DEFAULT }
ALTER ROLE { role_specification | ALL } [ IN DATABASE
database_name ]
  SET configuration_parameter FROM CURRENT
ALTER ROLE { role_specification | ALL } [ IN DATABASE
database_name ]
  RESET configuration_parameter
ALTER ROLE { role_specification | ALL } [ IN DATABASE
database_name ]
  RESET ALL
```

```
where role_specification can be:
    role_name
  | CURRENT_ROLE
  | CURRENT_USER
  | SESSION_USER
```

SET: https://www.postgresql.org/docs/current/sql-set.html

```
SET [ SESSION | LOCAL ] configuration_parameter { TO | = } { value
| 'value' | DEFAULT }
SET [ SESSION | LOCAL ] TIME ZONE { value | 'value' | LOCAL |
DEFAULT }
```

SHOW: https://www.postgresql.org/docs/current/sql-show.html

```
SHOW name
SHOW ALL
```

Configuration parameters

e.g. Basic access

```
> SHOW ALL;                                 -- list all
> SHOW search_path;                         -- specific

> SET search_path TO public,myschema;       -- set
```

5 levels of configurations:
- **system [lowest priority]**: system default
- **database**: per database
- **role**: per role
- **session**: per session
- **local [highest priority]**: per transaction

3 ways of setting configurations:

a) **postgresql.conf** file:
See below for details

b) **client CLI**:
e.g. use **PGOPTIONS**

```
$ PGOPTIONS='-c transaction_timeout=30s -c work_mem=40MB' psql ...

> SHOW transaction_timeout;
30s
```

c) **SQL**
5 levels:

ALTER SYSTEM SET ...	system level defaults Configs are saved in **<PGDATA>/postgresql.auto.conf** Effective only after **pg_reload_conf()**
ALTER DATABASE mydb SET ...	database defaults
ALTER ROLE yui SET ...	role defaults
SET ... **SET SESSION** ... **SET LOCAL** ...	current session

*** Values**

```
-- Boolean
on   off
true false
yes  no
1    0

-- String
abc
log          -- enum string
             -- e.g. client_min_messages = 'log'

-- Numeric
1
1.1
1.1MB        B|kB|MB|GB|TB
1.1s         us|ms|s|min|h|d
```

*** postgresql.conf**
Ref: https://www.postgresql.org/docs/current/config-setting.html#CONFIG-SETTING-CONFIGURATION-FILE

postgresql.conf
postgresql.auto.conf
Configuration file used by server i.e. **postgres** command
postreql.auto.conf contains configuration set dynamically by **ALTER SYSTEM**.
Location

- **<PGData>/postgresql.conf**
- **<PGData>/postgresql.auto.conf**

usage: check config file location
Use **config_file** parameter

```
> SHOW config_file;

                  config_file
-----------------------------------------------
 /opt/homebrew/var/postgresql@17/postgresql.conf
(1 row)
```

usage: see defined configurations
Uses **pg_file_settings** table

```
> select * from pg_file_settings ;
                      sourcefile                      | sourceline
| seqno |            name            |      setting       | applied
| error
------------------------------------------------------+------------
+-------+----------------------------+--------------------+--------
-+-------
 /opt/homebrew/var/postgresql@17/postgresql.conf      |         65
|     1 | max_connections            | 100                | t
|
 /opt/homebrew/var/postgresql@17/postgresql.conf      |        129
|     2 | shared_buffers             | 128MB              | t
|
 /opt/homebrew/var/postgresql@17/postgresql.conf      |        152
|     3 | dynamic_shared_memory_type | posix              | t
|
 ...
 /opt/homebrew/var/postgresql@17/postgresql.auto.conf |          3
|    14 | transaction_timeout        | 32s                | t
|
 /opt/homebrew/var/postgresql@17/postgresql.auto.conf |          4
|    15 | client_min_messages        | log                | t
|
 ...
```

```
# Comments

# Basic
search_path = public                    -- single
search_path = 'public', '$user'         -- multiple

# Boolean-like
row_security = on
row_security = off
row_security = yes
row_security = no

-- String
search_path = public
search_path = 'public'    -- same
```

```
-- Numeric
work_mem = 8000000
work_mem = 8MB            -- with units
                          -- B|kB|MB|GB|TB
work_mem = '8MB'          -- same
transaction_timeout = 1s  --


-- Enum (string)
client_min_messages = log
client_min_messages = 'log'

-- Numeric
```

*** SHOW/SET**

e.g.

```
-- Database level
-- NOTE: effective next session
> ALTER DATABASE mydb SET transaction_timeout='30s';


-- Role level
-- NOTE: effective next session
> ALTER ROLE yui SET transaction_timeout='30s';


-- Session level
> SET transaction_timeout='30s';


-- Transaction level
> SET LOCAL transaction_timeout='30s';
```

```
-- Show effective config
SHOW search_path;


-- Basic SET syntax
--
SET search_path=public;
SET search_path TO public;
SET search_path=DEFAULT;                    -- reset to default
SET search_path=public,pg_catalog;          -- multiple
SET search_path='public','pg_catalog';      -- multiple with quotes
SET search_path='public,pg_catalog';        -- BAD
```

```
-- String
SET search_path=public
SET search_path="public"

-- Boolean
SET row_security=on;
SET row_security=off;
SET row_security=TRUE;
SET row_security=FALSE;

-- Enum
SET client_min_messages=log;
SET client_min_messages='log';

-- Numeric
SET work_mem=80000;
SET work_mem ='8MB';       -- with units
SET work_mem =8MB;            -- ERROR
```

*** usage: ALTER SYTEM SET**

For setting **system** level configs

System config

```
> ALTER SYSTEM SET transaction_timeout='30s';

-- Restart
> SELECT pg_reload_conf();
```

<PGDATA>/**postgresql.auto.conf**

```
# Do not edit this file manually!
# It will be overwritten by the ALTER SYSTEM command.
transaction_timeout = '30s'
```

*** usage: find effective configuration parameter**

pg_settings: https://www.postgresql.org/docs/current/view-pg-settings.html

Use **pg_settings** view

source	default | configuration file | database | role | client | session The current effective source

e.g. see effective value of search_path

```
>  select * from pg_settings where name='search_path' \gx
-[ RECORD
1 ]---+-------------------------------------------------------------------
------
```

```
name            | search_path
setting         | "$user", public
unit            |
category        | Client Connection Defaults / Statement Behavior
short_desc      | Sets the schema search order for names that are
not schema-qualified.
extra_desc      |
context         | user
vartype         | string
source          | default
min_val         |
max_val         |
enumvals        |
boot_val        | "$user", public
reset_val       | "$user", public
sourcefile      |
sourceline      |
pending_restart | f

> set search_path to 'pg_catalog,public';

>  select * from pg_settings where name='search_path' \gx
-[ RECORD
1 ]--+-----------------------------------------------------------------
---------
name            | search_path
setting         | "pg_catalog,public"
unit            |
category        | Client Connection Defaults / Statement Behavior
short_desc      | Sets the schema search order for names that are
not schema-qualified.
extra_desc      |
context         | user
vartype         | string
source          | session
min_val         |
max_val         |
enumvals        |
boot_val        | "$user", public
reset_val       | "$user", public
sourcefile      |
sourceline      |
pending_restart | f
```

==========

==Configuration parameters==

*** Basic**

Ref: https://www.postgresql.org/docs/current/runtime-config.html

```
# File directories
data_directory                    e.g. /opt/homebrew/var/
postgresql/
config_file                       e.g. /opt/homebrew/var/
postgresql/postgres
hba_file
ident_file
external_pid_file


## Connection parameters
#
listen_address=localhost              TCP address
port=5432
unix_socket_directories=/tmp          e.g. /tmp/.s.PGSQL.5432
... etc


## Resource parameters
#
shared_buffers=128MB
work_mem=4MB                          memory used by single query
hashed_mem_multiplier=2.0
maintenance_work_mem=...
...


## client
search_path='$user',public
statement_timeout=0                       0=disabled
transaction_timeout=0                     0=disabled
default_transaction_read_only=false       whether a new
transaction is readonly by default
transaction_read_only=false               whether current
transaction is readonly
...



# configurations
autovacuum=on
autovacuum_max_workers=3
autovacuum_naptime=1min
autovacuum_vacuum_threshold=50
autovacuum_vacuum_scale_factor=0.1        10%

autovacuum_vacuum_insert_threshold=
autovacuum_vacuum_insert_scale_factor=

autovacuum_analyze_threshold=0.1
autovacuum_analyze_scale_factor=50
```

```
autovacuum_freeze_max_age= 200000000                    200 million
autovacuum_multixact_freeze_max_age= 400000000          400 million

autovacuum_vacuum_cost_delay=2ms
autovacuum_vacuum_cost_limit=-1
```

*** Log**

```
log_destination=stderr                                  stderr|csvlog|
jsonlog   log method
log_destination=stderr,csvlog                           multiple

# File logging
logging_collector=off                                   on|off
on=enable file logging
log_directory=log
log_filename=postgresql-%Y-%m-%d_%H%M%S.log
...

# Syslog
syslog_facility=LOCAL0
syslog_ident=postgres
...

#
log_min_message=WARNING
DEBUG5|...DEBUG1|INFO|NOTICE|LOG|WARNING|ERROR|FATAL|PANIC
                                                        minimum level
to log
log_min_error_statement=ERROR                           minimum error
leve
...
```

*** Performance Configurations**
https://www.postgresql.org/docs/current/runtime-config-query.html

```
-- geqo
 geqo='on'               boolean

geqo_threshold=12    number of JOINs to switch to geqo


-- query tuning          Default
enable_bitmapscan        TRUE
enable_indexscan         TRUE
enable_seqscan           TRUE

enable_nestloop          TRUE
enable_hashjoin          TRUE
enable_mergejoin         TRUE
... etc ...
```

```
-- options                    Default
from_collapse_limit           8 https://www.postgresql.org/docs/
current/explicit-joins.html
join_collapse_limit           8
...
```